Fiscal Year:	FY 2008	**Task Last Updated:**	FY 09/25/2007
PI Name:	Freed, Lisa E M.D., Ph.D.		
Project Title:	Microgravity Tissue Engineering		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
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No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 (jvp 6/2009) NOTE: Received NCE to 11/30/2008 (from 11/30/2007) per P. Denkins/JSC (7/07) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	Prof. Laura Niklason is no longer working on this project since she has left Duke University.		
COI Name (Institution):	Vunjak-Novakovic, Gordana (Columbia University and M.I.T.) Rupnick, Maria (Harvard Medical School) Guilak, Farshid (Duke University Medical Center) Padera, Robert (Harvard Medical School)		
Grant/Contract No.:	NNJ04HC72G		
Performance Goal No.:	1.21		
Performance Goal Text:	Sustain a leading biotechnology research program that will assure continued scientific and technical leadership.		

Task Description:	Our research involves cell- and tissue-levels studies of 3D engineered tissue constructs and utilizes models of relevance to tissue loss in astronauts in microgravity and skeletal and myocardial tissue regeneration in human patients on Earth. We selected to study skeletal and cardiac tissue regeneration because: (a) spaceflight adversely affects these tissues, (b) improved treatments are needed for millions of patients who suffer from progressive skeletal and cardiac diseases, (c) these two distinctly different tissue types can provide paradigms for studies of individual and interactive effects of biophysical stimuli, and (d) we and others have already demonstrated the feasibility of engineering skeletal and cardiac-like tissue, and thus can capitalize on recent advances to promote groundbreaking research and technology transfer. We previously designed in vitro models based on cells, 3D biomaterial scaffolds, and advanced bioreactor systems, and showed bioreactor-grown constructs were larger and functionally superior to controls grown in petri dishes. The current limitations of our tissue engineering research are as follows: (i) long periods of in vitro culture are required to generate mechanically functional engineered cartilage constructs, (ii) cardiomyocyte viability, differentiation, and contractility are poor in engineered cardiac constructs, and (iii) spaceflight tissue culture studies are limited by the lack of proper controls necessary to derive underlying mechanisms. We are addressing these limitations by: (a) integrated studies in cell and molecular biology, mass and momentum transport, design and operation of bioreactors including microgravity analog systems, tissue physiology and biomechanics, and implant biocompatibility and remodeling, (b) experiments interpreted with the aid of mathematical models, (c) complementary in vitro and in vivo studies, and (d) functional evaluation at various hierarchical scales, i.e. molecular, cell, and tissue. The project is organized as two Aims: (1) skeletal tissue engineering, and (2) cardiac tissue engineering. The global hypothesis is that biophysical stimuli that are physiologic in nature, magnitude and regime of application can direct cells to regenerate functional tissues in vitro. We can test this hypothesis in controlled studies of cells cultured on 3D scaffolds in bioreactors and in the presence of biophysical stimuli. The work is expected to help define how biophysical cues regulate skeletal and cardiac tissue development and function, and yield engineered skeletal and cardiac tissues with superior functionality compared to previous reports. Our studies of skeletal and cardiac tissue constructs involve the cultivation of cells on 3D scaffolds in advanced bioreactors and in the presence of biophysical stimuli. Critical research areas include cell sourcing (cell expansion, stem cell differentiation), development of novel biomaterial scaffolds and advanced bioreactors for cell seeding of scaffolds and construct cultivation in the presence of biophysical stimuli (e.g. controllable hydrodynamic shear, cyclic stretch, growth factors, pharmaceutical agents), and construct implantation in vivo for assessment of cell survival and graft integration, and capacity for defect repair. Tissue engineered constructs are assessed with respect to structure, molecular composition, and mechanical and electrophysiological function in order to evaluate individual and interactive effects of experimental variables. Key equipment includes facilities for working with cells, tissue specimens, experimental animals, polymer processing, bioreactor systems, biomechanical and electrophysiological test systems, and analytical instruments for histology, biochemistry, molecular biology, microscopy and image analysis. Construct properties are correlated with in vitro culture conditions and the outcome of in vivo studies, in order to provide feedback necessary to tailor model system parameters and obtain the best overall result. The expected outcomes are improved understanding of how biophysical stimuli regulate skeletal and cardiac tissue development and function, and the generation of tissue engineered constructs with functionality superior to that previously reported in the literature.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research is relevant to the Bioastronautics Roadmap (BR) questions #2) impaired fracture healing, #3) injury to joints, #5) occurrence of serious cardiac dysrhythmias, #6) diminished cardiac function, #19) pharmacology of space medicine, and possibly also to #21) rehabilitation on Mars. Our combined use of 3D engineered tissue constructs and advanced bioreactor systems provides a unique model system for controlled in vitro studies at the cell and tissue-levels that may be relevant to treating skeletal and myocardial tissue loss and dysfunctions in astronauts. Biophysical stimuli play a critical role in tissue growth, maintenance, and remodeling. During human spaceflight, altered biophysical stimuli and the resulting systemic adaptations can cause fluid and electrolyte imbalance, altered neurohormonal regulation, diminished tissue remodeling, and loss of bone and muscle mass. Exercise regimens and pharmaceutical interventions have been proposed to try to maintain skeletal and cardiac tissues in-flight, and to try to rehabilitate deconditioned skeletal and cardiac tissues post-flight. However, the ability to determine the underlying mechanism(s) or assess the potential risks/benefits of currently used countermeasures is limited by a lack of experimental models. Our controlled in vitro studies of how 3D tissue engineered constructs respond to biophysical stimuli are expected to help identify how the microgravity environment of space may compromise the ability of skeletal and cardiac tissues to respond to and/or withstand biophysical stimuli. Improved understanding can reasonably be expected to aid in the development of better countermeasures. Our tissue engineering research has implications not only for astronauts but also for humans on earth, e.g. by developing in vitro analogs for experimental pharmaceutical studies, and new treatments for skeletal and cardiovascular diseases. Skeletal and cardiac tissues perform structural and mechanical functions that are vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over 40 million Americans suffer from arthritis, which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited capacity for self-repair, and no currently available therapy can predictably restore a functional articular surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle, and heart disease is the leading cause of morbidity and mortality in developed countries. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting 3D tissue constructs that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, despite many advances, tissue engineered grafts for cartilage and cardiac regeneration still face many challenges in order to be of widespread clinical use, hence, this tissue engineering research can clearly benefit life on Earth. To date, this NASA research project has been highly productive, yielding more than 90 publications in leading peer-reviewed scientific journals including the one-and-only long duration microgravity tissue engineering experiment carried out aboard STS-79, Mir Space Station and STS-81, and published by Freed et al., Proceedings of the National Academy of Sciences USA. Also, this NASA research project resulted in 125 abstracts and five patent applications, and has helped to train sixteen postdoctors, nine graduate students including six who earned PhD degrees, and numerous undergraduates. Professional activities of the co-investigators include teaching of undergraduate and graduate students (please see the preceding paragraph), chairing and organizing scientific conferences, and serving on scientific research panels and scientific and editorial advisory boards. NASA used the data and experience gained in our ground and flight studies to develop two bioreactor systems: (i) Advanced Rotating Bioreactor System and (ii) Cell Culture Unit. The

Co-investigators use research supported by this NASA project as examples in the courses that we teach. For example, L.E. Freed has taught graduate-level courses in the Harvard-MIT Division of Health Sciences and Technology, e.g. "Biomaterials and Tissue Engineering for Medical Devices" (HST-521), and "Principles and Practice of Tissue Engineering" (HST-535), and F. Guilak has taught courses at Duke University and the Center for Biomolecular and Tissue Engineering., e.g. "Stem Cell Biology" (MCB/CBI-208), "Introduction to Biologically Inspired Materials" (MEMS-265/BME-265), the Rheumatology Basic Science Series, and the Osteoarthritis Series. Hence, this research has had a major impact on America with Earth benefits.

Task Progress:

The current approach to cartilage tissue engineering wherein cells and 3D scaffolds are combined to create implantable grafts is limited by failure of existing scaffolds to mimic the complex mechanical and structural properties of native articular cartilage. Biomaterial scaffolds provide one important means of controlling the architecture and mechanical properties of an engineered tissue [Freed et al., Tissue Engineering, 2006]. Hence, we designed and fabricated a biomimetic scaffold by three dimensional weaving of biocompatible fibers [Moutos et al., Nature Materials, 2007]. In particular, a computer-controlled microscale weaving technique was utilized to create anisotropic 3D woven structures that were then infiltrated with a chondrocyte-hydrogel mixture. The resulting engineered cartilage constructs exhibited approximately three orders of magnitude difference in tensile and compressive moduli, aggregate moduli of ~0.2 MPa and tensile moduli of ~300 MPa, therein mimicking the anisotropy, viscoelasticity, and tension-compression nonlinearity of native articular cartilage. Cartilage implants based on this 3D microwoven scaffold have the potential to provide mechanical function immediately after cell seeding without requiring prolonged cultivation in vitro. Moreover, the large number of variables in this scaffold design, e.g. selection of fiber material, number of fibers up to several hundred, density and packing in all directions, provides a tremendous range of possible mechanical properties. Previous cartilage tissue engineering studies in which cells and 3D scaffolds were combined and cultured in conventional petri dishes reported inefficient and inhomogeneous cell seeding of the scaffold, and slow regeneration of functional cartilage. Perfusion bioreactors, i.e. devices that can provide convective-diffusive mass transport, can enhance the efficacy of cell seeding and functional 3D tissue regeneration. However, previously developed perfusion bioreactors were limited and cumbersome because they requiring two distinct system components (one to provide gas exchange and another to provide perfusion) and, often, two distinct culture devices (one for cell seeding of scaffolds and a second for tissue regeneration). Hence, we developed a novel oscillatory perfusion bioreactor suitable that provided effective mass transport of gases, nutrients, and regulatory molecules in the context of a single, integrated and commercially applicable device [Moretti et al., Provisional Patent Application, 2007]. Significant improvements in chondrocyte seeding of 3D scaffolds and rapid regeneration of cartilage implants were demonstrated in the new bioreactor as compared to conventional in vitro culture methods. In particular, the new bioreactor significantly increased the DNA (cell) content of 4-day constructs and yielded histologically mature cartilaginous constructs within a 2-week period, in stark contrast to constructs that were seeded and cultured in petri dishes. These results suggest the bioreactor can provide high efficiencies of cell seeding and mass transport, thereby inducing rapid cartilage formation [Moretti et al., 2007 Abstract].

Tissue engineered cardiac constructs have been proposed as a platform for systematic in vitro studies of complex stimuli that occur in vivo, and as a means to repair damaged myocardium. However, the clinical use of engineered cardiac constructs is currently limited by suboptimal cardiomyocyte survival, differentiation, and contractility. We showed that the supplemental growth factors insulin-like growth factor-I (IGF), insulin, platelet-derived growth factor (PDGF), and angiopoietin (ANG) enhanced these properties for constructs made from heart cells and collagen sponges [Cheng et al., Tissue Engineering, 2007]. Both IGF and insulin significantly improved the construct contractile amplitude and excitation threshold as compared to unsupplemented controls, and IGF also significantly increased the amount of cardiac troponin-I and enhanced cell viability by three different assays (MTT, LDH and TUNEL). PDGF significantly increased construct contractile amplitude compared to controls, and enhanced cell viability by all three aforementioned assays and ANG enhanced cell viability by LDH assay. These findings show that properties of cardiac tissue constructs are differentially enhanced by supplemental growth factors, and imply that engineered cardiac constructs do provide a platform for systematic in vitro studies that can potentially identify mechanisms for enhancing myocardial regeneration.

Supplemental IGF utilized in concert with perfusion culture of constructs in our new oscillatory perfused bioreactor further enhanced the formation of engineered cardiac constructs [Cheng et al., 2007 Abstract and 2008 Circulation Suppl.]. In particular, constructs made from heart cells and collagen sponges were cultured in either the new bioreactor or conventional petri dishes using medium supplemented or not with IGF. Perfusion culture significantly enhanced heart cell survival and contractile activity as compared to non-perfused controls, as assessed by MTT, TUNEL and electrophysiological assays. Heart cell elongation and myofibrillogenesis were readily observed in perfusion culture, in stark contrast to non-perfused controls, as assessed by immunostaining for cardiac troponin-I. Effects of supplemental IGF and perfusion culture were independent, additive, and highly significant ($p<0.0001$ for contractile amplitude; $p<0.005$ for excitation threshold), suggesting at least partially different cell activation mechanisms. Moreover, spontaneous construct contractility was present only in perfusion culture and not in petri dishes. These findings demonstrate that the new oscillatory perfused bioreactor and supplemental IGF can improve the viability, differentiation, and contractility of cardiomyocytes in engineered cardiac grafts.

The current approach to cardiac tissue engineering is limited by failure of existing scaffolds to mimic the complex mechanical and structural properties of native myocardium. For example, collagen sponges, although sufficiently compliant to allow for contractility, fail to promote native-like cell alignment due to isotropic structure and generally insufficient mechanical strength to support high tensile loading. We designed and fabricated new scaffolds with controlled mechanical anisotropy and cell-orienting pore structures by combining excimer laser microablation and lamination technologies [Engelmayr et al., 2007 Abstract]. Microfabrication of an elastomeric biomaterial yielded scaffolds with highly controlled, anisotropic pore designs that exhibited anisotropic mechanical properties similar to native ventricular myocardium. Moreover, the anisotropic scaffolds with cultured neonatal rat heart cells yielded engineered cardiac constructs that exhibited directed contractility and significantly higher index of cellular orientation compared to scaffolds with isotropic pore designs. These findings show that scaffolds microfabricated to include anisotropic pore designs can overcome the principal structural-mechanical limitations of previous scaffolds to enhance the properties of engineered cardiac grafts.

Heart failure is characterized by fibrosis that is thought to occur via processes involving matrix metalloproteinases (MMPs). We hypothesized that the cellular composition of engineered cardiac grafts would mediate their remodeling potential by a mechanism involving MMPs, and tested this hypothesis by comparing 3D cultures of heart cells enriched

	for cardiomyocytes (the CM group) with co-cultures of CM and cardiac fibroblasts (the CM-CF group) in media supplemented or not with the MMP inhibitor doxycycline [Nichol et al., 2007 Abstract]. The CM-CF group exhibited a higher degree of cell-to-cell network formation and cell orientation than the CM group, and doxycycline eliminated network formation and cell orientation in both groups. These findings were associated with the following differences in MMP expression: (i) the CM-CF group exhibited higher amounts of total and active MMP-2 than the CM group, (ii) the CM group exhibited significantly higher amounts of total and active MMP-9, than the CM-CF group, and (iii) doxycycline reduced the amounts of all MMPs in both groups. These findings show that the cellular composition of engineered cardiac grafts can influence their remodeling and potential for fibrosis via MMP-mediated mechanisms. During the current reporting period, our research was featured on the covers of two leading scientific journals (Nature Materials and Tissue Engineering), a documentary film entitled “The New Frontiers of Life”, and in NASA Tech Briefs (volume 31, number 9, article MSC-23647). Overall Task Progress during the current reporting period includes 7 conference presentations, 1 published abstract, 5 published and in-press scientific research articles (4 other articles are currently in preparation), 2 published book chapters, and 1 new provisional patent application (see Section 6).
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